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Deep Learning Approaches for Classifying Informal and Formal English Texts Using Linguistic Features

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ABSTRACT Effective techniques for automatically classifying texts are becoming increasingly necessary due to the exponential expansion of digital material. Differentiating between formal and informal documents can help students identify appropriate resources for their assignments and improve the effectiveness of information retrieval systems. Although machine learning is extensively utilized in classification of text, there is a lack of research focused to the effective differentiation of formal and informal writings through linguistic features. This gap highlights the necessity for advanced methodologies that improve classification accuracy and enhance the value of digital content in academic and retrieval systems. Our research addresses the problem by utilizing deep learning methodologies and a wide range of 13 linguistic attributes to get enhanced efficacy in text classification. Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and Long Short-Term Memory Networks (LSTM) were considered. A dataset , including both formal (news articles, formal documents) and informal (personal letters, personal blogs) texts, were gathered from several web sources. We considered linguistic markers such as colloquialisms, contractions, modal verbs, slang, acronyms, pronouns, phrasal verbs, grammar complexity, vocabulary complexity, voice, and language type to generate the feature vector. The feature vectors were utilized to train and assess the classification models using several cross-validation techniques, particularly 3, 5, 7, and 10 folds. The efficacy of the models was evaluated using performance indicators, f-measure, accuracy, precision, and recall. With the highest accuracy of 99.8% and resilience in differentiating between formal and informal texts, the LSTM model outperformed than the others. Future research will examine big datasets, more linguistic characteristics, sophisticated deep learning models, and real-time and multilingual classification systems.

INDEX TERMS ANN; CNN; Document Classification; Formal Documents; Informal Documents; LSTM

I. INTRODUCTION

In the current highly competitive global environment, students and researchers have the internet as their main source of educational information, thus navigating a web of resources. It has also changed the way instructions are given and provided access to a vast amount of instructional information. As important reference sources, educational websites offer a variety of materials including academic papers, journal articles, educational manuals, and current event updates. The Ken showed the how important of using formal writing in academic materials [1]. The authors who write for these environments can use either the formal or the informal writing style.

In the field of education, researchers and students employ various papers to enhance their knowledge, gain understanding, and obtain data. These resources include a variety of text types and the learners have to choose the most appropriate one at any given time. Thus, the ability to distinguish between formal and informal writing is crucial to the process of learning. The importance of the distinction between different writing styles is becoming more important and is highlighted in the example described. The contrast between these examples highlights the importance of knowing the audience and how to use specific language for the audience. The ability to recognize and understand the differences between formal and informal language and understand how to use them appropriately has become more important as digital technologies and the internet develop. If it could be automated whether the document is formal or informal, it would make the students' task easier and permit them to take out probably more concise the relevant information.

Currently, information retrieval is a laborious and time-consuming operation due to the unorganized nature of web resources. Agnes represented from his research how formal and informal language act through the internet [2]. As of today, information retrieval is a laborious and time-consuming process due to the unorganized nature of web resources. A possible solution is to apply document classification, which is the practice of labelling previously unlabelled documents. This method increases extracting the useful information from various large number of digital resources. This automatic classification system can be used to dramatically improve the access and use of educational resources by students. Ultimately, this will improve students' success rates in academics.

The objective of our study is to automatically classify documents as either formal or informal style using language features distinctive of the respective styles. We aim to construct a model that helps learners make distinctions between formal and informal documents, enabling the learner to more effectively choose the appropriate resources for their individual learning needs.

For this study computational linguistic gives the significant support. Computational linguistics involves the utilization of computer technology for the study and understanding of both formal and informal language. Computational Linguistics is an academic domain that integrates linguistics, computer science, and artificial

intelligence to analyse language from a computational perspective [3].

Recent studies highlight that incorporating linguistic features such as syntax, sentiment, and morphology improves text classification accuracy. Research on psychological state classification demonstrates the effectiveness of linguistic feature integration [4]. A 2024 survey on computational morphology underscores its role in enhancing classification models [5]. Additionally, a comparative study found that combining neural networks with linguistic features leads to more precise results [6] , reinforcing the importance of linguistic analysis in style-based classification.

In the field of Computational Linguistics, the classification of texts by style requires linguistic component analysis. Specific decisions on syntax and word and phrase choice affect style, according to Karlgren. The use of vocabulary becomes an indicator of style; for instance, long, complex sentences are typically interpreted as being formal, but sentences that incorporate informal idiomatic expressions are considered to be informal most often. In addition, numerous additional factors influence whether a piece of writing is formal or informal, and formal and informal writing has a wide range of sub-genres. It is essential to first determine the primary traits that set formal writing apart from informal writing to create an efficient classification model. These features include variations in vocabulary utilization overall as well as word choice, phrases, and expressions. First of all, we need to define what are the main characteristics that distinguish formal writing from informal writing to obtain an effective classification model. These characteristics are reflected in overall vocabulary usage as well as specific word choices, phrases, and expressions. By capturing these we can build a model that can consistently classify documents and hence provide students with a tool for navigating the vast amount of material offered. This characteristic will enable students to more easily and efficiently access the most relevant and suitable information needed for their learning tasks saving also time.

A. Formal and informal writing style

From the formal and informal language that uses the most precise, objective, and specific terminology one can refer to a formal style. This type of language is used in academic books and articles, technical reports, research papers as well as legal documents. Formal style finds in scholarly writing and professional documentation the kind of expression required by accuracy, clarity, and attention to detail.

An informal style of writing, or speaking, is a friendly, warm approach that is used to conduct writing or in everyday conversation. It is employed when writing for or conveying to close friends, relatives, etc, and in general conversation. This sort of style may be used in personal emails and face-to-face conversations.

The following table 1 show the key features of formal and informal writing styles. Here we identified important 13 linguistic features with referencing [7] and [8].

Table 1. Key features and examples for formal and informal writing styles

Features	Formal	Informal
Colloquialisms [7]	Do not use colloquialisms Ex: Want to	Use colloquialisms Ex: Wanna
Contractions [7]	Do not use contractions Ex: Can't	Use contractions Ex: Can not
Pronouns [7]	Third person pronouns Ex: This could be good research	First-person pronouns Ex: I think this good research
Phrasal verbs [7]	Do not use phrasal verbs Ex: Spend time	Use phrasal verbs Ex: Hang out
Grammar [8]	Complex grammar Ex: Global warming, which results in several major environmental concerns, is one of the most prevalent environmental challenges.	Simple grammar Ex: Global warming is an environmental challenge.
Modal Verbs [8]	Used modal verbs Ex: This methodology can be applied to the research	Do not use modal verbs Ex: This methodology used for the research
Vocabulary [8]	Ex: Purchase	Ex- Buy
Abbreviations [8]	Do not use abbreviations Ex: Examination	Use abbreviations Ex: Exam
Acronyms [7]	Do not use acronyms Ex: As soon as possible	Use acronyms Ex: ASAP
Slangs [7]	Do not use slang Ex: Friend	Use slangs Ex: Mate
Initialism [7]	Do not use initialisms Ex: United Kingdom	Use initialism Ex; UK
Language [8]	Formulaic language Ex: The experiment's result showed the validity of the accuracy.	Direct language Ex: My parents are very well.
Voice [9]	Passive Voice Ex: dissertations are written by scholars.	Active Voice Ex: Scholars write dissertations

B. Research questions and objectives

Figure 1 shows the mapping between research questions and objectives.

Distinguishing between formal and informal writing styles is crucial for learning and understanding. This study aims to automatically classify documents as formal or informal using above mentioned 13 language features distinctive of each style. After collecting formal and informal documents, we applied different pre-processing techniques and then extracted the features. Then, research evaluates and compares different deep learning models such as

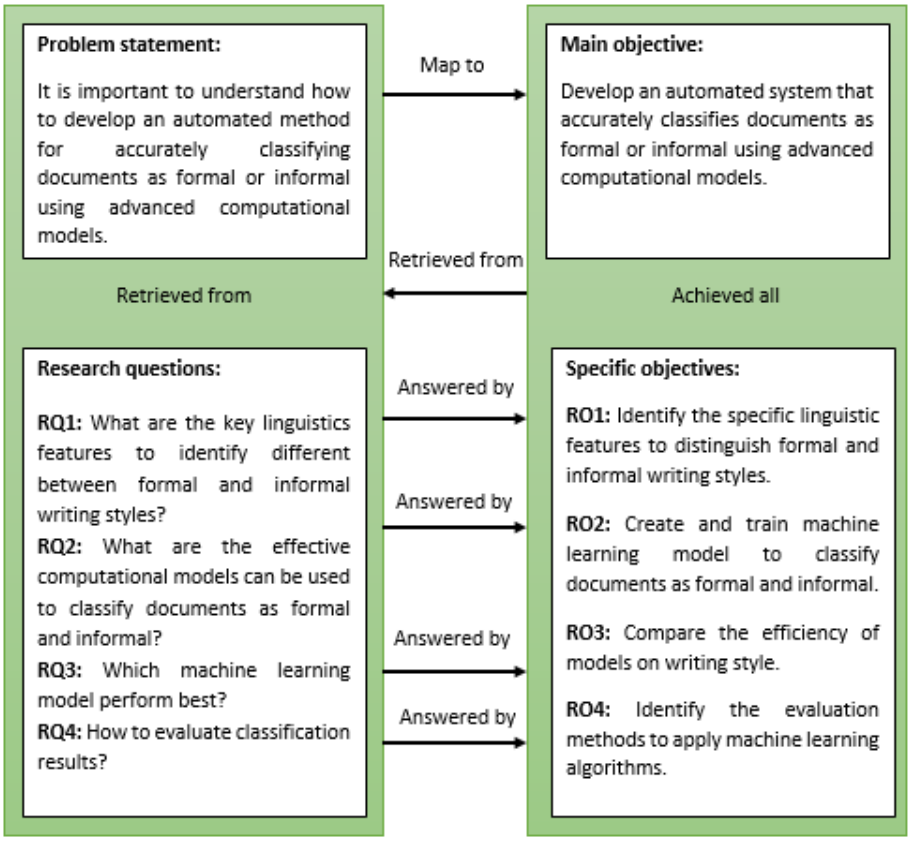

Fig. 1.Mapping between research questions and objectives

ANN, CNN, and LSTM to identify the best classification method. The evaluation is performed based on accuracy, precision, recall, and f-measure to better understand how these models are in classifying texts based on their writing style. This will ensure that the performance of models will be systematically analysed to choose the best model for document classification tasks between formal and informal texts.

The rest of this study is organized as follows: Related works on writing style-based document classification are investigated in Section 2. In Section 3 we present a novel automated mechanism for labelling documents in formal and informal style. The studies conducted on the effectiveness of our method, as well as the metrics and measurement methods applied, are detailed in Section 4.

Section 5 concludes the paper, summarizes findings, and proposes future research directions.

II. LITERATURE REVIEW

A. Related work

In the literature review, we look at previous studies and approaches to the problem of classifying documents according to writing style, with a particular emphasis on attempts to automate the differentiation between formal and informal styles by linguistic and computational methods.

Classes of Hindi poetry were identified by [10] as part of their suggested categorization scheme: Shringar, Karuna, and Veera. They generated their feature vectors using the Bag of Words model. Five machine learning classification techniques were classified: Support Vector Machines (SVM), Random Forest, Decision Tree, K-nearest Neighbors (KNN), and Naïve Bayes. Tests revealed that in this instance, the accuracies of Random Forest and Naïve Bayes outperformed those of the other algorithms. A university website was utilized as a case study in [11] and the Waikato Environment for Knowledge Analysis (WEKA) machine learning workbench served as the framework for the machine learning effort. The classification model was created using the Naïve Bayes method. The study's findings demonstrated that the Naïve Bayes algorithm is capable of accurately classifying massive online content. On the other hand, the researchers employed a single classification method. Various feature vector types were covered by [12] to represent and subsequently classify documents. Regarding their impact on vector and document classification, the study contrasted the Binary, Count, and Term Frequency–Inverse Document Frequency (TF–IDF) feature vectorsThey employed the Naïve Bayes classification for every feature vector representation. Once more, just one classification method was utilized in this study. In [13]examined many categorizations of methods, including Decision Tree, SVM, and Naïve Bayes, in their survey study. Additionally, they examined assessment metrics that may be used for different types of study, including accuracy, F-measure, and G-measure. They only examined three classification methods, though. Using the Naïve Bayes methodology, [14]) developed a classification system for Turkish papers. F-measure, recall, and accuracy were used to gauge performance. Once more, the model in this study was constructed using a single classification method. A research paper classification method that can group research articles into relevant groupings based on related issues was suggested [15]. The TF-IDF values of each

paper were utilized to classify whole papers using the K-mean clustering approach. Keywords in the abstracts were represented using the Latent Dirichlet Allocation technique. Here, the drawback was that the research publications were classified using just one classification system.

The WEKA text mining technology was utilized by [16] to assist in the classification of online frauds. The classification models were constructed using the J48 Decision Tree, Sequential Minimal Optimization, and Naïve Bayes techniques. They discovered that J48 produced the best results among these techniques in terms of accuracy and error rate. The authors of [17] presented a neural network design that demonstrated how effectively both compression and iteration layers could encode character input. They evaluated the suggested model on eight extensive document classification tasks and contrasted it with a character-level convolution-only model.

In eight languages, a new subset of the Reuters Corpus with balanced class priors was proposed by [18]. In terms of phrases and morphology, they studied four extremely diverse languages: Italian, Russian, Japanese, and Chinese. Multilingual word and phrase embedding in all language translation directions, provide a solid basis. The Bidirectional Encoder Representations from Transformers (BERT) model was initially used to document categorization [19]. They demonstrated how BERT may be used to create a basic classification model for four widely used datasets.

In [20] presented a novel neural network-based model that measures uncertainty using a newly developed dropout entropy technique. A state of uncertainty concerning the appropriate course of action is referred to as uncertainty. Additionally, they presented a metric learning system based on dropout approaches that enhances feature representation for uncertainty. In their accurate prediction studies, this exhibited negligible predictive variability. Specifically, their model's accuracy increased from 0.78% to 0.92% when 30% of the most ambiguous results were applied to the "20 Newsgroups" dataset used by anthropologists. A two-stage text-document classification system that combined automated and conventional feature engineering was presented by [21]. The suggested process consisted of deep component modification using filters in conjunction with a neural network selection algorithm. The "20 Newsgroups" dataset and the BBC News dataset, two of the most popular public datasets, were utilized to assess this technique. The findings of the evaluation indicate that the suggested methodology surpasses the current state-of-the-art approaches that rely on deep learning and machine learning. For "20 Newsgroups," the margins were 7.7%, while for BBC News, they were 6.6%. Note that neural networks were the main focus of this study.

In [11] used the LitCovid database, an expanding collection of 23,000 research publications on the new 2019 Coronavirus, to conduct a study of alternative multiple-label document classification methods. By a narrow margin, they discovered that the pre-trained language models outperformed all other baseline models on this dataset, with BioBERT and micro F1 achieving accuracy scores of around 86% and 75% of the test set, respectively. In further research, [22] examined the processing of documents from different time intervals with document classification systems that were trained on documents from certain intervals, taking into account both seasonal and non-seasonal intervals. In [23] introduced a notably intricate document representation format that employs in-depth learning to automatically classify financial documents into predetermined categories. The two primary components of their model architecture were document classification and representation. Their results demonstrated that the best document classification performance for the INFUSE dataset was obtained by a hidden three-layer nutrient network consisting of 1024 neurons. In [24] concentrated on classifying information using KNN and Naïve Bayes, two of the six primary techniques. The TREC Code of Conduct, which can be downloaded and contains over 3,000 text documents with over 20 categories, served as the corpus for this study. After processing the data with RapidMiner, k=13 was found to be the ideal value for k in the KNN. This number for k resulted in an average accuracy of 55.17%, which was higher than Naïve Bayes' 39.01%. We do see, though, that they only employed two classification techniques.

In [25] used a majority vote to classify assignments using five well-known classification techniques for the Urdu language corpus. 21,769 news articles across seven categories could be found in the corpus. The papers were pre-processed data using tokenization, stop word removal, and a rule-based stemmer because the algorithms were unable to operate directly on the data. Following pre-processing, the data has 93,400 characteristics taken out. With a majority vote, the outcomes had a 94% recall and accuracy rate. Lemmatization and lower-case conversion were not used as pre-processing methods in this study. The goal of the study [26] was to use a Naïve Bayes classification algorithm to estimate a song's singer based only on the lyrics. The 207 songs that Metallica and Nirvana played were included in the dataset that was developed. With an F-measure of 0.94, a recall of 0.95, and an accuracy of 0.93, the model assessment procedures produced extremely positive results. As a result, words classified with Naïve Bayes were deemed successful. This study, however, only employed one classification method.

Machine learning and automated document classification approaches are discussed in [27] article, which lessens the workload for domain experts by managing a huge number of news items or web page information. The authors spoke about a variety of design choices and factors that often come up while creating automatic classifiers with the help of XML document structure. Using two datasets, experimental research on automated classification has been conducted in this work. New texts were classified using the learned Naïve Bayes classifier. The outcomes were contrasted using a hierarchical classifier and a KNN. Real-time web apps may employ this kind of capability because of the incredibly optimistic categorization performance demonstrated by these trials.

The paper [28] addressed an important issue in computational linguistics: distinguishing between formal

and informal styles of text in document classification and text generation. They proposed two main techniques to solve this task namely building a model to classify any text or sentences and a second technique based on natural language generation (NLG). They started with summarizing characteristics of formal and informal writing styles and manually collected parallel lists of formal and informal words. Then they built the model using a Decision Tree, Naïve Bayes, and a Support Vector Machine (SVM). The evaluation results showed that that model can predict a class and formal and informal for any text or sentence with high accuracy. Then they built a system that can generate formal and informal sentences by using NLG techniques. However, they did not focus on every characteristic of formal and informal writing styles as well as they used only three machine-learning models for the experiments. The study done by, [29] aimed to work with machine learning techniques to deliver a model that can distinguish formal and informal texts. They explored the Decision Tree, Random Forest, and Logistic Regression as machine learning algorithms. As a result, they found that features are crucial in text formality

classification. As well as their study included how to transform informal text into formal text and vice versa. In this study, they experimented with only five key features to identify the difference between formal and informal writing styles and also used only machine learning algorithms.The research article [30] proposed a document classification method based on formal and informal styles. That research used 200 text documents and used the tf-idf feature. In the classification process Decision Tree (J48), Random Forest, Multilayer Perception (MLP), and SVM as machine learning models. The result was investigated using 5-fold cross-validation and the Random Forest algorithm showed the highest accuracy. This experiment used a few data sets, and because of this, there can be problems with accuracy. The paper [31] presented a method to identify automatically the style of a particular

document. For the feature extraction used tf-idf and five classification algorithms with an ensemble learning approach. From them, Random Forest showed the highest accuracy but after being compared with the Ensemble learning algorithm it represented the highest accuracy. As the results they investigated a higher accuracy can be obtained by adding several algorithms. In this study not used any deep learning algorithms to combine with other algorithms.

The research article [1] proposed a method to identify the academic documents and as the methodology they used Tree tagger. According to their finding it was essential to identify academic documents for the educational purposes. The paper [2] presented usage of formal and informal language for the communication. They used qualitative method for investigations. As the finding they showed that for mobile communication most probably used the informal language.

Recent studies have advanced the classification of formal and informal English texts by integrating deep learning techniques with linguistic features. [32] conducted a systematic study comparing statistical, neural-based, and Transformer-based methods for formality detection, highlighting the effectiveness of neural networks in capturing linguistic nuances. Additionally, [33] study proposed a concept for using linguistic knowledge to support binary text classification, emphasizing the importance of linguistic-driven feature selection in enhancing model performance. Furthermore, [34] study introduced deep learning models, such as Formality-LSTM and Formality-BERT, for formality prediction

Table 2. Summary of literature

Paper	Paper purpose	Methodology	Limitations
[1]	How used formal writing in academic writing	Tree Tager	Not used the deep learning algorithms
[2]	Usage of formal and informal language for communication	Qualitative method	Not used qualitative method to get the accuracy. Not used deep learning algorithms
[10]	Classification approach for Hindi poetry	Random forest SVM Decision Tree KNN Naïve Bayes	Not using any deep learning algorithms.
[28]	Distinguish between formal and informal styles of text in document classification and text generation.	Decision Tree Naïve Bayes SVM	Focused only on eight characteristic Used only three machine learning models for the experiments.
[29]	Work with machine learning techniques to deliver a model that can distinguish formal and informal texts.	Decision Tree Random Forest Logistic Regression	Experimented with only five key features. Used only three machine learning models for the experiments.
[30]	Document classification method based on formal and informal styles.	Decision Tree Random Forest MLP, SVM	Used a few data sets. Not specify what are the characteristics used.
[31]	Identify automatically the style of a particular document.	Decision Tree Random Forest MLP, SVM, Naïve Bayes Ensemble learning approach	Not using any deep learning algorithms. Not discussed features separately.

without the need for feature engineering, demonstrating the potential of deep learning approaches in this domain.

These studies collectively underscore the pivotal role of combining deep learning models with linguistic features to effectively classify texts based on formality.

B. Research gap

Despite significant progress in the field of computational linguistics, automatic document style classification, that is, the identification of formal and informal writing styles, is still a difficult and emerging field of study. However, the studied literature provides certain helpful insights and outlines the general approaches and methods that need further research due to the existing gaps and limitations.

As for prior studies, they have successfully investigated language features that are essential for text formality determination; however, the scope of their findings has often been limited to several features. It is essential to address a gap in the literature by compiling 13 different linguistic features into a single feature vector. Some of these characteristics are subjectivity and objectivity markers, use of slang, number of words in a sentence, number of formal and informal words used, use of contraction, number of passive voices used, etc. By systematically dealing with these processes, classification models improve their stability and accuracy and thus provide a more accurate classification of text formality. The most frequently used algorithms include Decision Tree, Naïve Bayes, Random Forest, SVM, and other similar methods. The use of complex deep learning techniques such as CNN, LSTM, and ANN networks has not been explored in detail, although there is potential in these approaches. The lack of research on deep learning-based solutions due to obstacles including the requirement for substantial processing resources and the availability of large, high-quality datasets, which are frequently limited. The intricate nature of implementation and the restricted interpretability of deep learning models could discourage academics from investigating these methodologies. Conventional methods, due to their greater accessibility and interpretability, are frequently favored across several domains. But it is possible that deep learning methods may be more efficient in detecting complex patterns in textual data. It is also possible to increase the classification accuracy by using deep learning algorithms.

Our research bridges important knowledge gaps through the implementation of 14 linguistic elements to construct an extensive feature selection method for document classification. Detailed analysis through this approach produces better distinctions between formal texts and informal texts in classification tasks. The comparison of advanced deep learning models ANN, CNN and LSTM for specific performance stands as our unique contribution which provides essential insights into their effectiveness. The combined elements in our study represent a major breakthrough for computational linguistics and text classification research direction.

Some of the studies involved a small number of documents in the dataset, which may have affected the results' reliability. Expanding the size of the dataset to include more documents could improve the performance of the classification algorithms.

Our work attempts to address these research gaps by developing a better classification model that involves several language features, advanced deep learning algorithms, and comprehensive evaluation metrics. It might significantly enhance the effectiveness of resources and information in educational and other digital content systems. Table 2 showed the limitations of the previous research as well.

III. METHODOLOGY

The following Figure 2 represents the proposed approach for the study.

A. Data Collection

In the present research, we selected several 5000 text files from several reliable web sources. These data are considered secondary because these data were originally collected for other purposes on the Internet. A suitable number of formal and informal text documents are incorporated into the collection.

We gathered new articles and some formal documents from the Kaggle website [35] and the Washington Post news website [36] to represent formal documents. These articles are excellent examples of formal writing because of their organized framework, formal tone, and respect for journalistic norms. We gathered personal letters and personal blogs for informal documents from the following websites: Writing Help Central [37], AnswerShark [38], Letters Free [39] and extra documents from Kaggle [35]. These personal letters are great examples of informal writing since they are usually written in a more relaxed and conversational tone.

The dataset only included text files in English to maintain linguistic consistency. Using the criterion of writing style and origin, every document was pre selected as formal or informal. This process of selection and classification helped us generate a large number of documents that are representative and diversified enough to be used for testing and training our classification algorithms. Table 3 showed some examples for formal and informal writing styles.

Table 3. Examples of formal and informal writing styles

Formal Writing Style	**Informal Writing Style**
The project's budget has been updated in light of current financial modifications.	We've changed the project's budget because of some recent financial adjustments
After the week, we will offer suggestions after thoroughly reviewing the plan.	Right now, we're looking over the proposal and will get back to you by the end of the week

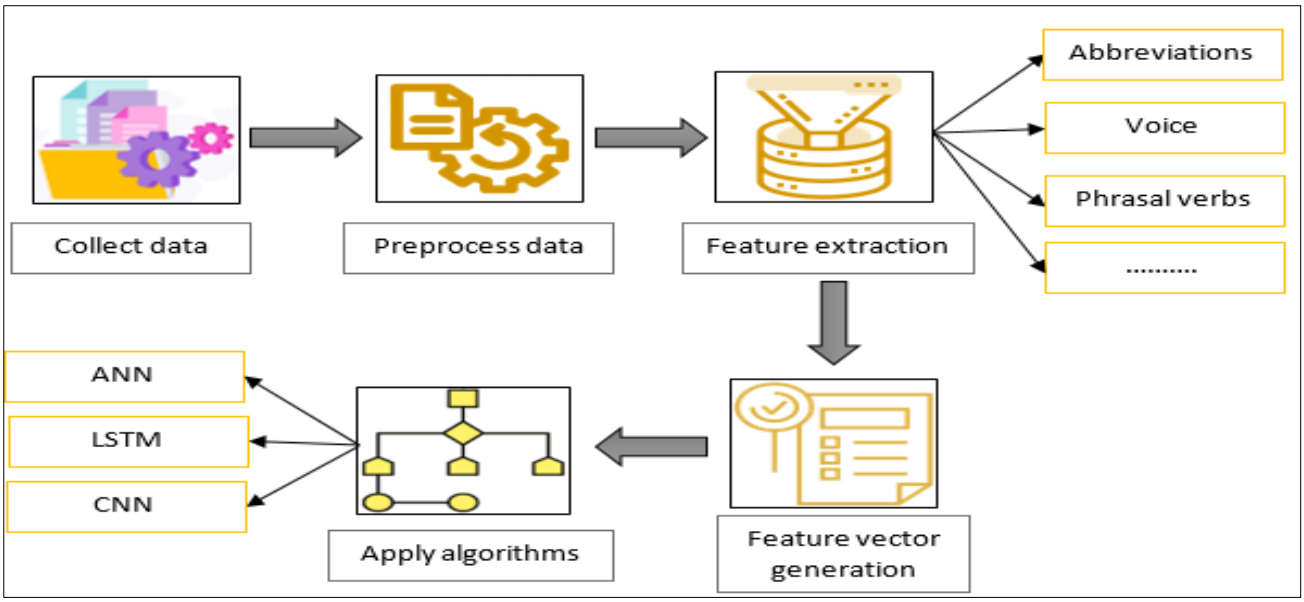

Fig. 2. Proposed Approach

B. Pre-process Data

Text preprocessing mainly involves the process of splitting the text file into words and removing unwanted elements. Four essential preprocessing procedures were involved: This includes lemmatization, lower-casing, stop word removal, and tokenization. These are important processes that help to clean and normalize the text data in a format that is suitable for further analysis and classification.

Tokenization: Tokenization is a process of segmentation of a text sequence into words, phrases, symbols, or other meaningful parts called tokens. This is a very important phase in text processing because it divides the text into segments that can be analysed separately, this is important for further stages.

Lower casing: Before the classification, all capital characters should be converted to lowercase. This phase ensures that all the text data is in the same feature space by projecting every word in the text file. For instance, after lower casing the strings "Education" and "education" as well as "EDUCATION" would be considered similar since it eliminates the differences in case sensitivity.

Stop word removal: To focus on the more informative and content-related words that are more relevant in distinguishing between the formal and informal texts, the stop words are removed before classification. They are words such as 'and', 'the', 'is', etc. which are normally insignificant when used in a categorization context. They have the capability of skewing the data and therefore compromise the efficiency of the classification process.

Lemmatization: Lemmatization is a technique of natural language processing that removes the affixes and prefixes from the words to bring them back to their stem. Lemmatization considers the context and returns the actual base form of the word. One might lower the terms "running," "ran," and “runs” to their stem, “run”.We were certain that the unprocessed text data was well-pre-processed by following the above-mentioned preprocessing steps. It will help enhance the current classification models to be able to differentiate between the formal and informal writing styles better.

C. Feature Extraction

One of the most important steps in converting pre-processed text input into measurable metrics that machine learning algorithms may use is feature extraction. We retrieved features that capture 14 different linguistic aspects indicative of formal and informal writing styles from our dataset of 5,000 text files. The following is a description of the characteristics and how they were extracted. Using algorithms developed in the Python programming language, the 14 distinct linguistic elements representative of formal and informal writing styles were extracted. In particular, we used libraries for natural language processing tasks like spaCy and NLTK. We converted our pre-processed text data into measurable metrics that machine learning algorithms could evaluate and use by applying these algorithms to them. The 13 distinct linguistic elements are as follows:

Colloquialism: The phrases and expressions used in colloquialism. These are the kinds of terminology used in informal writing. For every document, we numbered and recognized. First, we gathered all colloquialisms through the internet and applied the algorithm to count the colloquialisms in our data set.

Abbreviation: In informal communications, abbreviations are more frequently utilized. Each document's usage of abbreviations was counted. Gathered a list of abbreviations through the internet and according to that identified the abbreviations in every document using an algorithm.

Contraction: Writing that uses contractions is considered informal. We tallied how many contractions each text

included (e.g., "don't," "can't"). We created seven rules to identify the contractions of the dataset. Rules are:

- "Rule 1": "(?:\w+)'m"
- "Rule 2": "(?:\w+)'s"
- "Rule 3": "(?:\w+)'re"
- "Rule 4": "(?:\w+)'ve"
- "Rule 5": "(?:\w+)'d"
- "Rule 6": "(?:\w+)'ll"
- "Rule 7": "(?:\w+)'t"

Modal Verbs: Probably, formal writings employ modal verbs. For every document, the frequency of modal verbs was determined. The word list is "shall", "should be", "can", "could", "will", "would", "may", "must", and "might". Those words are identified in the data set.

Slang: Slang is an effective means to determine whether writing is informal. A list of slang phrases was compiled with the sample slang gathered from the internet.

Acronyms: Both formal and informal situations employ acronyms, however they may not be used as frequently. Each document's total number of acronyms was tallied reference acronyms gathered from the internet.

Initialism: Every document's initialization frequency (e.g., "FBI," "ASAP") was tallied. Though they are also used informally, initializations are frequently used in formal settings.

Pronouns: First, identify the first person and third person pronouns and then according to that separately take it as two characteristics.

First-Person Pronouns: How many first-person pronouns (such as "I," or "we") were used? In informal writing, first-person pronouns are more frequently used.

Third-person pronouns: (such as "he," "she," and "they") were also counted. In formal writing, third-person pronouns are frequently employed.

Phrasal Verbs: Two rules were created for identifying the phrasal verbs in the dataset as follows and separately take it as two characteristics.

Verb + Adverb: Phrasal verbs, which are verbs followed by an adverb (such as "give up"), were recognized and tallied.

Verb + Preposition: Phrasal verbs, which are verbs followed by a preposition (such as "look into"), were also counted. Phrasal verbs are usually more prevalent in writing that is informal.

Grammar: Define a set of dependency labels that indicate complex structures of the sentences in a dataset. The following two features separately take as two characteristics.

Complex Grammar: A count was conducted to determine the frequency of complex grammatical structures, such as compound-complex phrases. Formal compositions tend to use more complex syntax.

Simple Grammar: The usage of basic grammatical constructions, such as simple phrases, was also counted. Writing that is informal frequently uses simple grammar.

Vocabulary: The percentage of complex phrases in each text was calculated to determine the vocabulary's complexity. Considering words with more than six letter as complex. More complicated language in writing is a sign of formal writing.

Voice: By creating the rule identify the active and passive voice sentences in every document. We will look for the 'VB' (verb base form) or 'VBD' (past tense verb) followed by 'VBN' (past participle verb) in the tagged words. If we find this pattern and 'BY' is not present in the sentence, we consider it passive and both are considered separately as two characteristics

Active Voice: We tallied how many times each document used an active voice structure. Informal writing tends to use the active voice more frequently.

Passive Voice: The instances of passive voice compositions were also tallied. In formal writing, the passive voice is generally more prevalent.

Language: First, count the frequency of each token. Then calculate the total number of tokens. According to that calculate the frequency of common phrases in the sentence as well as calculate the ratio of common phrases to total tokens. After that classify the language based on the ratio. If it is formulaic language return 0, direct language return 1, and for the mixed language return 2.

This allowed us to transform the textual data into quantitative representations that reflect the nature of

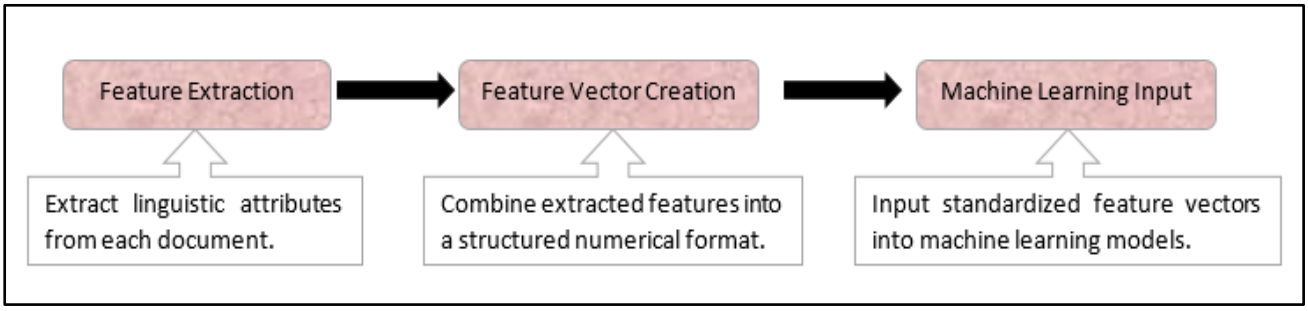

Fig.3. Process of feature vector generation

Colloquialism	Abbreviation	Contraction	F/S Pronouns	Third Pronouns	Phrasal verbs (verb+adverb)	Phrasal verbs (verb+preposition)	Simple Grammar	Complex Grammar	Modal Verbs	Vocabulary (Complexity %)	Slangs	Active Voice	Passive Voice	Acronyms	Initialism	Language Formulaic Language =0 Direct Language =1 Mixed Language =2	Target Variable
0	63	0	0	0	2	1	4	5	2	33	4	8	1	5	1	1	0
0	66	0	0	0	0	2	4	6	1	30	0	9	1	3	1	1	0
1	70	0	0	0	1	0	3	12	0	39	0	12	3	1	1	1	0
0	55	2	0	0	2	0	5	6	3	43	0	10	1	0	3	1	0
1	73	1	0	0	0	1	9	9	1	33	3	15	3	2	0	1	1
1	72	0	0	0	0	0	3	12	1	40	1	12	3	2	2	1	1

Fig. 4 Sample for feature vector generation

formal and informal writing with the help of these characteristics extracted. These attributes are fed into our machine learning models which in turn enables them to learn and differentiate between different writing styles.

D. Feature vector generation

As shown in Figure 3, feature extraction is the first step of transformation which is used to transform the text data into feature vectors that are used by machine learning models.

In this step, linguistic features are detected and counted in each document. Linguistic features can be, for example, contractions, abbreviations, or colloquialisms. In the next step, i.e., feature vector creation all detected linguistic features are put in predefined structured numerical representation. The same representation is used for all documents to ensure the standardized form final numeric representation that is given as an input to machine learning models which analyze generated patterns and characteristics to classify writing styles and increase the predictive capabilities of machine learning models.

Figure 4 shows the sample for feature vector generation. Here explain the different features (characteristics) in different columns as mentioned above and the last column is for two target variables as formal (1) or informal document (0). Structuring the data in this manner ensures that our methodology properly captures the distinctive aspects of formal and informal documents using all 13 linguistic features. This generated feature vector provides to the classification models and it enhances the overall efficacy of our methodology.

E. Apply algorithms

The study utilized ANN, CNN, and LSTM algorithms to analyse the dataset. ANN handles complex pattern recognition, CNN excels in feature extraction, particularly for grid-like data, and LSTM is effective for sequential data due to its ability to capture long-term dependencies.

ANN: Each layer of an ANN is composed of several neurons or perceptron. An ANN only processes inputs in a forward direction and therefore is also called a feed-forward neural network [40]. Figure 5 explains the architecture of ANN algorithm

CNN: Today CNNs are one of the most popular models for deep learning. This neural network computational model includes one or more layers of convolution that can be either pooled or fully connected. It is based on a type of multilayer perceptron [40]. Figure 6 showed the architecture of CNN algorithm.

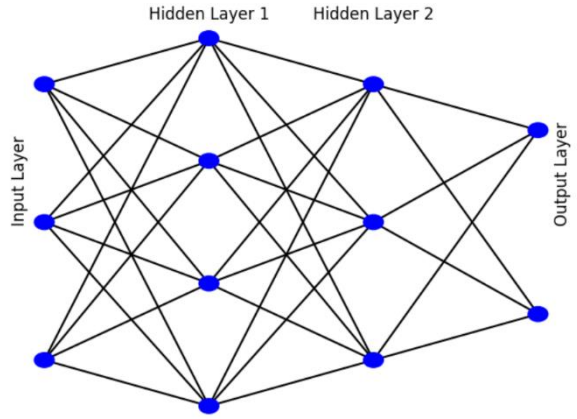

Figure 5. Ann Architecture

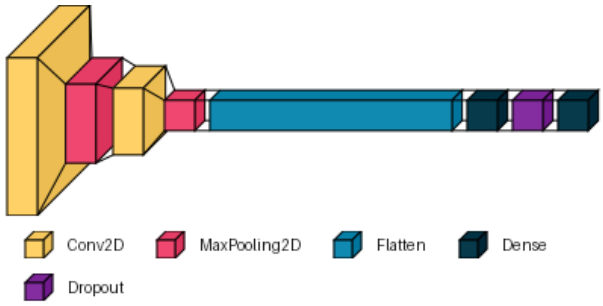

Figure 6. CNN Architecture

LSTM: In the LSTM network, the memory cell is regulated by the input, forget, and output gates. These gates define the input, output, and the addition of data in this memory cell. Therefore, LSTM networks can learn long-term dependencies and the information that flows through the network is either retained or rejected [41]. LSTM model showed by following Figure 7.

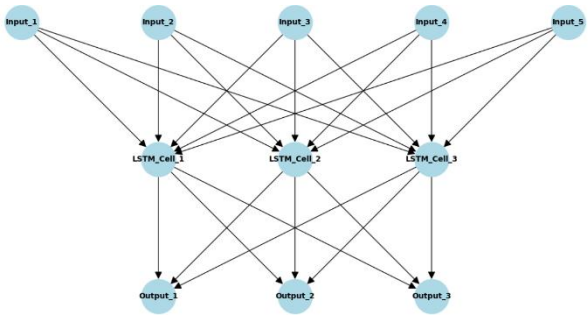

Figure 7. LSTM Architecture

Table 4. Confusion matrix

	5 Fold				10 Fold				3 Fold				7 Fold			
	TP	**TN**	**FP**	**FN**	**TP**	**TN**	**FP**	**FN**	**TP**	**TN**	**FP**	**FN**	**TP**	**TN**	**FP**	**FN**
ANN	411	437	88	64	198	234	40	28	693	724	142	107	294	324	55	41
LSTM	588	392	8	12	291	194	6	9	960	639	26	41	425	282	3	5
CNN	449	427	73	51	234	201	36	29	766	680	123	98	337	292	58	28

IV. RESULTS AND DISCUSSION

The experimental platform used Microsoft Windows 10 on a PC with Processor Intel® Core (TM) i5-8250U CPU @ 1.60GHz, RAM 4.0GB.

For this study used Python programming language and Jypyter Notebook as the platform with the cross-validation namely 5 fold, 10 fold, 7 fold, and 3 fold. These metrics have been calculated for test dataset. The test dataset comprised 20% of the overall dataset, facilitating a comprehensive assessment of the models' efficacy in distinguishing between formal and informal texts.

A. Confusion matrix

The confusion matrices for the various cross-validation folds (5, 10, 3, and7) of the classification algorithms (ANN, CNN, and LSTM) offer a comprehensive. understanding of each model's performance in terms of

true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). This study was conducted using the data that was supplied as shown in table 4.

C. Accuracy of classification algorithm

In this study, we used several cross-validation folds (3-fold, 5-fold, 7-fold, and 10-fold) to assess the performance of three machine learning algorithms: ANN, CNN, and LSTM.

Figure 8 below provides an overview of each algorithm's

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

All folds consistently demonstrated that the LSTM model outperformed the other, achieving the highest accuracy of 99.8% in the 7-fold cross-validation. Even in the 3-fold cross-validation scenario, the LSTM model's lowest accuracy of 96.4% surpasses the maximum accuracies attained by the ANN and CNN models. This highlights the superiority of the LSTM model in our classification task due to its adeptness.

D. Results of precision, recall and f-measure

We used precision, recall, and f-measure across various cross-validation folds (3-fold, 5-fold, 7-fold, and 10-fold) assess the performance of the classification algorithms, which are ANN, CNN, and LSTM, in addition to accuracy. All three methods had their precision, recall, and F-measure computed; the computations for these metrics are detailed in the following (2), (3), and (4).

$$Precision = \frac{\mathrm{TP}}{TP + FP} \quad (2)$$

$$Recall = \frac{\mathrm{TP}}{TP + FN} \quad (3)$$

$$F - Measure = \frac{2 * Precision * Recall}{Precision + Recall} \quad (4)$$

Table 5 below provides a summary of the comprehensive result.

Table 5. Results of the performance matrix

		ANN	LSTM	CNN
3 Fold	**Precision**	0.85	0.97	0.87
	Recall	0.85	0.97	0.83
	F-measure	0.85	0.97	0.85
5 Fold	**Precision**	0.86	0.98	0.88
	Recall	0.86	0.98	0.84
	F-measure	0.86	0.98	0.86
7 Fold	**Precision**	0.86	1.00	0.88
	Recall	0.86	1.00	0.85
	F-measure	0.86	0.99	0.86
10 Fold	**Precision**	0.86	1.00	0.88
	Recall	0.86	0.97	0.84
	F-measure	0.86	0.98	0.86

All folds consistently demonstrated precise precision, recall, and f-measure values, highlighting the outstanding performance metrics of the LSTM model. The 7-fold cross-validation obtained the highest values, achieving an F-measure of 0.99, with recall and precision both reaching 1.00. The LSTM exhibited proficiency in distinguishing between formal and informal texts, showcasing remarkable metrics across 3-fold and 10-fold cross-validations.

D. Results of MAE and RMSE

Then, using the formulas in (5) and (6), we determined the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) for each of the six methods. In this case,

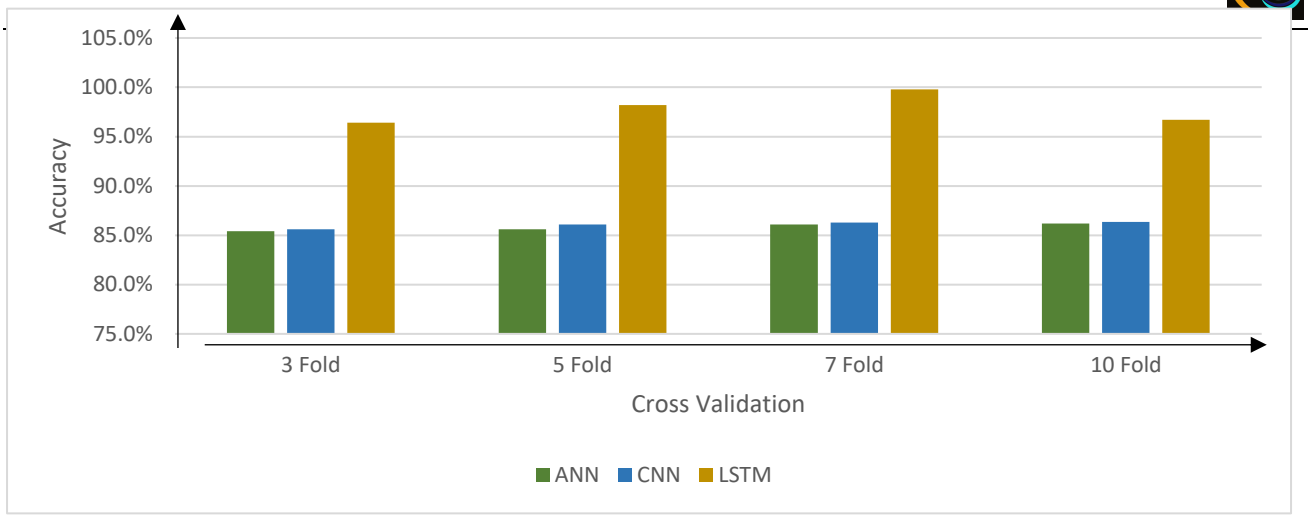

Fig 8. Accuracy of classification algorithms

Mvx is the expected result, Pvx is the current labeled value depending on the outcome, and T is the number of projected values.

$$MAE = \frac{1}{T}\sum_{x=1}^{T} | p_{vx} - M_{vx} | \quad (5)$$

$$RMSE = \sqrt{\frac{1}{T}\sum_{x=1}^{T}(p_{vx} - M_{vx,})^2} \quad (6)$$

To assess the models' performance, the MAE and RMSE was also computed, as shown in Figure 10 and 11.

The MAE values of both the ANN and LSTM models remained consistent across all folds, with the ANN slightly surpassing the LSTM in terms of error rate. Likewise, similar to the ANN model, the LSTM model showed steady RMSE values (ranging between 0.37 and 0.38) across each. The low and stable error rate of the LSTM model indicates reliable performance.

Furthermore, each classifier's hyperparameter and optimum values were identified as indicated in table 6. According to the results obtained from the ANN, LSTM and CNN showed better accuracies in epochs 200, 200, and 10 respectively. Difference between the LSTM model and the ANN and CNN models is manifested in the classification results of documents into formal and informal categories. The superior performance of LSTM in correctly identifying the true positive and true negative is evident concerning the false positive and false negative across all folds. However, the ANN, and CNN, were inferior to the LSTM in classification.

Table 6. Configuration parameter values

Configuration parameters for classification algorithms	
Algorithm	**Hyperparameter and Optimum value**
LSTM	Epochs = 200, batch size = 32, activation = “relu”, optimizer = ”adam”, verbose = 0
ANN	Epochs = 200, batch size = 32, activation = “relu”, solver = “adam”
CNN	Epochs = 10, batch size = 32, verbose = 0, activation = “relu”, optimizer = “adam”

E. Comparison with existing approach

Figure 9 compares the results of our study with previous ones. It's important to note that each study used different datasets, with all studies using datasets smaller than the 5000 data set we utilized.

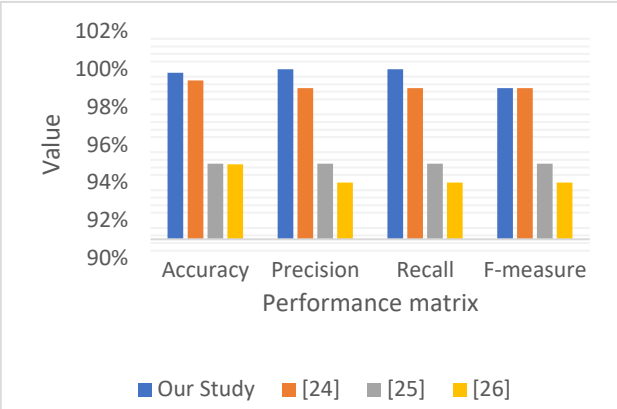

Figure 9. Comparison results with previous studies

In the study [24], SVM achieved the highest accuracy with a dataset of 1000. In the research [25], Random Forest yielded the highest values with a dataset of 1000, while the remaining study [26] used a dataset of 200, where SVM showed the highest performance. However the dataset and methodology are different, our proposed approach showed the better accuracy, precision, recall and f-measure than the existing approaches.

V. CONCLUSION

This is crucial in the recent digital age, where there are huge amounts of different text generated on numerous platforms, to identify and classify a given text in Formal or Informal ways. In the end, we aimed to establish effective methods for differentiating between these writing styles by exploiting and evaluating cutting-edge machine learning models and natural language processing techniques. We wanted to enhance the accuracy & efficiency of textclassification by focusing on an extensive feature extraction process and converting these features into some proper indexed numerical representation. This study provides a valuable methodology for computational linguistics. The study collected 5000 data sets, which were

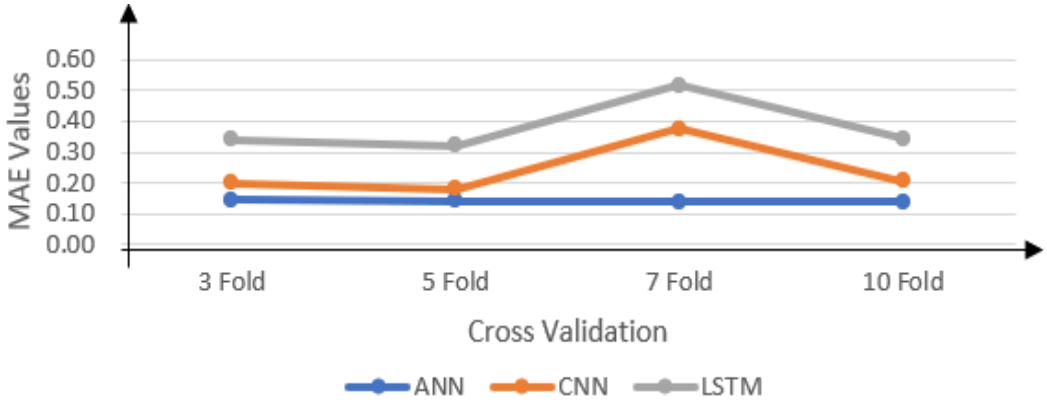

Fig 10. Results of MAE

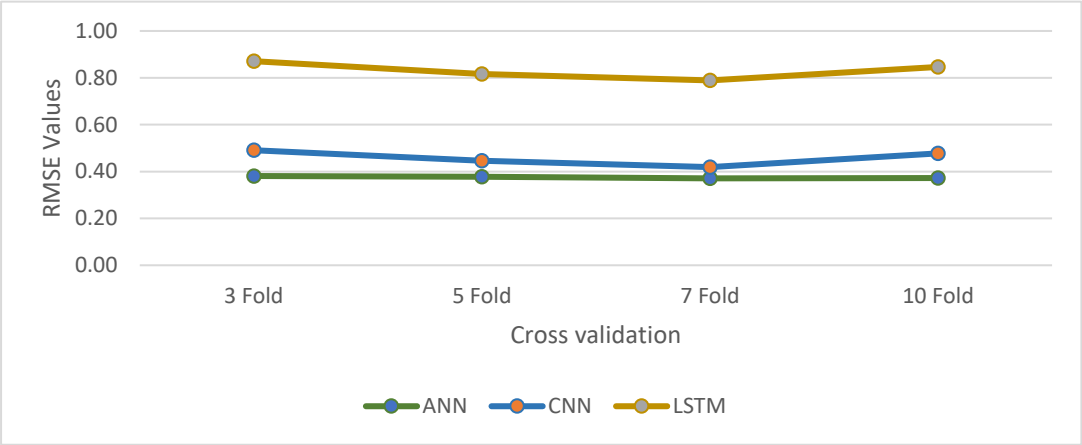

Fig 11. Results of RMSE

then pre-processed. Herein, the characteristic extraction process encompassed counts for colloquialisms, abbreviations, contractions, modal verbs, slangs, acronyms, initialisms, pronouns, phrasal verbs, grammar complexity, vocabulary complexity, voice and language type that gave a complete set of features which was very necessary for the classification task. These characteristics were transformed into machine learning models usable format during the feature vector creation process. We analyzed three deep-learning models to classify formal and informal documents. considered LSTM, CNN, and ANN. The models were used for their classification performance using metrics such as accuracy, precision, recall, f-measure, MAE, and RMSE over multiple cross-validation folds. Results showed that the LSTM model had a better performance than ANN or CNN in terms of accuracy, precision, recall, and f-measure. In particular, the LSTM performed better at reducing both false positives and negatives as well as achieving the highest accuracy (99.8%). To conclude, the outcomes of this study contribute to computational linguistics by giving valuable ways that distinguish between formal and informal writing styles through machine learning and deep learning techniques.

These findings are practical in improving recommendation algorithms for professional and educational writing tools.

A. Future Work

The future work involves collecting larger datasets from different domains and incorporating the additional linguistic features as well. As well as investigating advanced deep learning architecture improvements in accuracy. Last, but not least combining machine learning and deep learning models into a hybrid approach could leverage the strength of both techniques for better performance.

This study's findings, hold considerable potential for practical applications. For instance, the classification framework could be implemented into automated educational programs to help students identify between formal and informal writing styles, boosting their academic writing skills. Likewise, these models could improve writing helpers by delivering prompt feedback on text style, assuring suitability for diverse settings such as professional emails, academic papers, or personal correspondence. Such applications show the broader societal influence and practical importance of our study.

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